

Rovauto Garage Flow

Current backend flow after latest server changes - application, admin review, wallet activation, booking acceptance, Cashfree verification, and WhatsApp links.

Current Scope

Garage application remains a basic form. Garage images/documents are not required during application. Image uploads can be added later after approval from the garage dashboard using existing Cloudinary utilities.

Important Activation Rule

Admin approval only verifies the garage. The garage stays inactive until its wallet has at least Rs.1000 credited through a Cashfree-verified recharge.

1. High Level Arrow Flow

1	Garage owner submits basic partner application form -> backend saves it as PENDING.
2	Admin reviews application -> approve, deny, or request changes.
3	Deny/request changes -> status updates and email is sent to garage owner through Resend.
4	Approve -> garage owner user, garage record, and garage wallet are created.
5	Garage is verified but inactive -> isVerified = true, isActive = false, wallet balance = 0.
6	Garage owner creates wallet recharge order -> backend creates Cashfree order.
7	Garage completes Cashfree payment -> backend verifies order directly with Cashfree.
8	If Cashfree says PAID -> backend credits wallet. If balance >= Rs.1000 -> garage becomes active.
9	Customer creates booking -> backend broadcasts to nearby active and verified garages.
10	Garage accepts booking -> backend deducts platform fee from garage wallet and confirms booking.
11	Customer gets garage details, WhatsApp link, and Google Maps direction link.

2. Application And Admin Flow

Step	Actor	What Happens
1	Garage owner	Submits basic partner application using POST /api/garage/applications.
2	Backend	Creates GarageApplication with status PENDING. Duplicate open applications for same email/phone are blocked.
3	Admin	Lists and opens applications using /api/admin/garage-applications endpoints.
4A	Admin	Request changes -> status becomes CHANGES_REQUESTED and email is sent with admin note.
4B	Admin	Deny -> status becomes DENIED and email is sent with admin note.
4C	Admin	Approve -> creates/updates GARAGE_OWNER user, creates Garage, creates GarageWallet, links application to garage.
5	Backend	Approved garage starts as isVerified = true and isActive = false until wallet activation is complete.

Admin Endpoints

GET /api/admin/garage-applications, GET /api/admin/garage-applications/:applicationId, POST /approve, POST /request-changes, POST /deny. These routes are protected for ADMIN.

3. Garage Wallet And Cashfree Flow

Step	Actor	What Happens
1	Garage owner	Opens wallet and starts recharge. Minimum recharge order amount is Rs.1000.
2	Backend	Creates GarageWalletTransaction with status PENDING and creates Cashfree order.
3	Cashfree	Garage owner completes payment using Cashfree checkout/session.
4	Backend	Verify endpoint fetches order from Cashfree and checks order id, amount, and INR currency.
5A	Backend	If Cashfree order_status is PAID -> wallet is credited and transaction becomes SUCCESS.
5B	Backend	If order is FAILED/EXPIRED/TERMINATED -> transaction becomes FAILED. Wallet is not credited.
6	Backend	If garage is verified and wallet balance is at least Rs.1000 -> garage becomes active.

Security Point

Wallet balance is not trusted from frontend. Backend credits wallet only after fetching and verifying the Cashfree order server-side.

4. Customer Booking And Garage Acceptance Flow

Step	Actor	What Happens
1	Customer	Creates booking with vehicle, services, schedule/location, and notes.
2	Backend	Calculates service amount and platform/handling fee.
3	Backend	Broadcasts booking to nearby garages that are active and verified.
4	Garage owner	Views incoming booking request and accepts or rejects.
5	Backend	On accept, checks garage wallet balance before confirmation.
6	Backend	Deducts garage platform fee from GarageWallet and creates GARAGE_ACCEPT_FEE transaction.
7	Backend	Booking becomes CONFIRMED, selected request becomes ACCEPTED, other sent requests expire.
8	Customer	Booking success/details show garage details, WhatsApp link, and directions link.

5. Platform Fee Deduction

Service Amount	Garage Deduction	Notes
SOS	Rs.50	SOS acceptance charge.
Rs.300 - Rs.999	Rs.30	Same slab logic as customer platform/handling fee.
Rs.1000 - Rs.4999	Rs.99	Deducted from garage wallet on accept.
Rs.5000 - Rs.19999	Rs.249	Garage cannot accept if wallet balance is insufficient.
Rs.20000+	Rs.500	Transaction stored as GARAGE_ACCEPT_FEE.

Acceptance Rule

If the garage wallet does not have enough balance for the acceptance fee, the booking accept action fails and the booking is not confirmed by that garage.

6. WhatsApp Link Flow

Where	Backend Output	Details
Garage list/detail	garage.whatsappLink	Generated from garage.whatsappNo or garage.phone.
Nearby garage response	garage.whatsappLink	Useful if frontend wants direct contact display.
Garage request response	request.garage.whatsappLink	Available where garage object is returned.
Booking success	whatsappLink and booking.garage.whatsappLink	Customer can open WhatsApp after garage is accepted/assigned.

WhatsApp Format

Backend normalizes Indian 10-digit numbers into <https://wa.me/91XXXXXXXXXX>. No extra database field is required because it uses whatsappNo first, then phone as fallback.

7. What Is Ready Vs Later

Area	Status	Comment
Basic garage application	Ready	No images required in application form.
Admin approve/deny/request changes	Ready	Backend endpoints exist. Admin UI is still separate frontend work.
Email updates	Ready	Uses Resend when configured.
Garage wallet recharge	Ready	Cashfree order creation and verification are in backend.
Garage activation	Ready	Requires verified wallet balance $\geq$ Rs.1000.
Garage accept deduction	Ready	Deducts same platform fee slab from garage wallet.
WhatsApp link	Ready	Returned by backend in garage/booking responses.
Application image upload	Later	Not needed now. Can be added after approval/dashboard using Cloudinary.
Customer before/after service photos	Later	Not part of this backend change.

8. Key Backend Routes

Purpose	Route	Access
Submit garage application	POST /api/garage/applications	Public
List applications	GET /api/admin/garage-applications	ADMIN
Approve application	POST /api/admin/garage-applications/:applicationId/approve	ADMIN

Request changes	POST /api/admin/garage-applications/:applicationId/request-changes	ADMIN
Deny application	POST /api/admin/garage-applications/:applicationId/deny	ADMIN
Garage wallet	GET /api/garage/wallet	GARAGE_OWNER / ADMIN
Create recharge order	POST /api/garage/wallet/recharge/order	GARAGE_OWNER / ADMIN
Verify recharge	POST /api/garage/wallet/recharge/verify	GARAGE_OWNER / ADMIN

Generated for the current Rovauto codebase state. Keep this document updated if the frontend admin panel, garage dashboard, or image upload stage changes.